IT5514 / APPLIED MACHINE LEARNING

# Predicting Seasonal Paddy Production in Sri Lanka

Explainable Boosting Machine (EBM) and four benchmark models



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## 1. Problem Definition and Dataset

Paddy production is important for Sri Lanka's food supply and agricultural planning. Harvest amounts vary between districts and the Maha and Yala seasons due to cultivated area, rainfall, temperature and other conditions. This study estimates district-season paddy production (tonnes) and examines the factors behind the predictions.

The combined dataset uses paddy statistics attributed to the Department of Census and Statistics (DCS), Sri Lanka, and weather information attributed to Open-Meteo. It covers historical district-season observations from 2010 to 2023.

| Dataset detail | Description |
|---|---|
| **Records and coverage** | 660 records; 25 districts; Maha and Yala; 2010–2023 |
| **Target** | Production_t (tonnes of paddy production) |
| **Main features** | Sown area, past production, rainfall, temperature, district, season and other weather indicators |

### Dataset preparation

Exploratory Data Analysis (EDA) was used to examine seasonal and district patterns, missing values and relationships between features. Five missing previous-year production values were filled using training-data medians. District and season were encoded for the four benchmark models; EBM used categorical values directly. Scaling was not required for these tree-based methods.

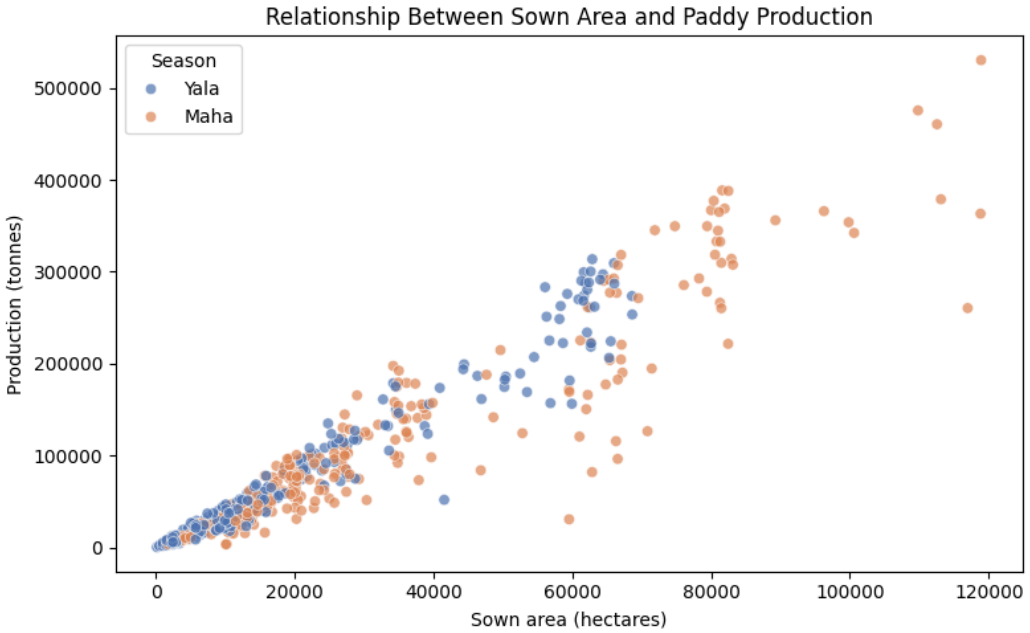



Figure 1. EDA: larger sown areas generally correspond to higher seasonal production.

## 2. Selection of Explainable Boosting Machine

Explainable Boosting Machine (EBM) was chosen because it learns nonlinear patterns while showing how individual features contribute to each prediction. It combines additive feature effects with boosting, making its results easier to interpret than many complex ensemble models. Random Forest, XGBoost, CatBoost and LightGBM were included as accuracy benchmarks.

## 3. Model Training and Evaluation

The data was divided by year instead of using a random split, so older records were used to predict newer ones. The unseen 2022–2023 data was reserved for the final test.

| Data split | Years | Purpose |
|---|---|---|
| Training | 2010–2019 | Initial model fitting |
| Validation | 2020–2021 | Initial comparison |
| Unseen test | 2022–2023 | Final evaluation |

**Hyperparameter tuning.** Three expanding validation periods were used: 2016–2017, 2018–2019 and 2020–2021, with only earlier data used for training. EBM's best settings were learning rate 0.02, max leaves 2 and two interactions.

MAE measures average prediction error in tones; RMSE gives more weight to large errors; $R^2$ shows how much variation in production the model explains. Lower MAE and RMSE are better.

| Model | Val. MAE | Test MAE | Test RMSE | Test $R^2$ |
|---|---|---|---|---|
| **Random Forest** | 12,477.69 | 23,945.11 | 41,249.95 | 0.76 |
| **XGBoost** | 11,439.53 | 25,882.50 | 42,527.50 | 0.75 |
| **CatBoost** | 12,461.23 | 25,742.12 | 44,288.15 | 0.73 |
| **LightGBM** | 11,602.43 | 26,220.02 | 43,684.44 | 0.73 |
| **EBM** | 19,310.48 | 27,667.97 | 43,429.50 | 0.74 |

Table 1. Five-model results. Validation MAE is averaged over three historical folds; all error values are in tonnes.

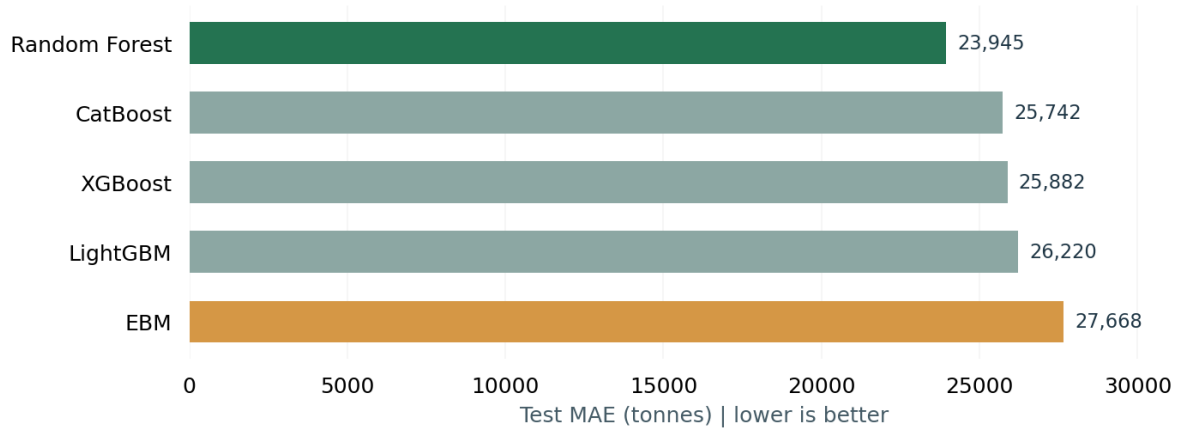



Figure 2. Mean absolute error on unseen 2022–2023 test records (lower is better).

XGBoost achieved the lowest validation MAE (11,439.53 tonnes), while Random Forest had the lowest observed test MAE (23,945.11 tonnes). EBM produced a test MAE of 27,667.97 tonnes and $R^2$ of approximately 0.74. It was less accurate than the benchmarks but offers built-in explanations of its predictions.

## 4. Explainability and Interpretation

Explainability was used to identify which inputs influenced predictions and to understand the reasons behind individual results.

### XGBoost — SHAP and LIME

**SHAP.** Sown area was the most important feature for XGBoost (average absolute SHAP impact about 73,525 tonnes). Seasonal maximum temperature and year had smaller effects. This suggests that the size of land cultivated strongly influences predicted total production.

**LIME.** For one prediction, sown area was again the strongest positive factor (about +171,275 tonnes), while more than six very hot days reduced the local explanation by about 5,996 tonnes. LIME's local fit $R^2$ was 0.75, so its explanation was useful but approximate.

### EBM — built-in global and local explanations

EBM can directly show which terms matter most across the dataset (global explanation) and how they add together for a particular prediction (local explanation).

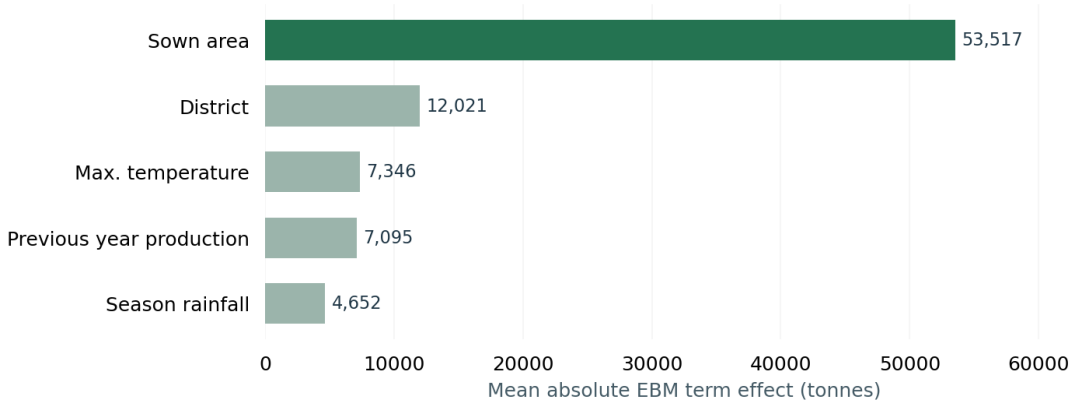



Figure 3. Most influential features in the EBM model (average absolute model contribution).

In one example for Ampara during Maha 2022, EBM predicted about 394,730 tonnes compared with actual production of 221,529 tonnes. Sown area increased the estimate the most, while previous-year production reduced it. This example also shows that an understandable prediction can still have a large error.

## 5. Discussion

**Data quality.** Only five values were missing, but the exact source tables and weather aggregation steps need clearer documentation. Predictions may change if source figures or seasonal definitions are revised.

**Model limitations.** Sown area is strongly connected to total production, so the models may rely more on land area than weather effects. EBM had noticeably higher errors in 2022 and the Maha season. Its Ampara example also shows that an individual prediction can be substantially wrong.

**Bias and fairness.** Error levels may differ between districts. A model that performs well on average could still be unreliable for particular farming areas, so district-level results should be checked before real-world use.

**Real-world use and ethics.** Because the inputs include observed weather over the full cultivation season, this is mainly an end-of-season production estimator, not a reliable before-planting forecast. Predictions should assist human planning, not replace official agricultural assessments.

**Possible improvements.** Use more recent years, verify the exact data sources, compare error by district, and test whether an early-season version can work using only information available at prediction time.

## 6. Conclusion

All five models were trained and compared on historical paddy production data. XGBoost performed best during validation and Random Forest had the lowest observed error on the final test period. EBM produced reasonable test results and clearer feature-level explanations, especially highlighting the importance of sown area.

### References

DCS Sri Lanka. Paddy Statistics: statistics.gov.lk/Agriculture/StaticalInformation/PaddyStatistics

Open-Meteo. Historical Weather API: open-meteo.com/en/docs/historical-weather-api

InterpretML. Explainable Boosting Machine: interpret.ml/docs/ebm.html